1. **Stakeholder analysis and conflict resolution**

Identification and interests

- **Principal:** Interests include institutional simplicity, low technical overhead for staff, and strict adherence to grade moderation policies to protect academic integrity.
- **Lectures:** Interests include immediate release of marks to students upon completion of marking and the flexibility to grant individual deadline extensions.
- **Head of Department (HoD):** Interests include oversight of all department marks, strict moderation before marks are released and the enforcement of consistent late penalty rules (deduct 10% per day for up to 3 days).
- **Student:** Interests include transparency, a reasonable resubmission policy before deadline, constructive written feedback rather than bare numbers and avoiding outright rejection of late work.
- **IT Officer:** Interests include offline functionality due to unreliable internet, old hardware performance capabilities and robust password or data protection.

Conflicts, Ambiguities and Resolutions

- **Mark release timing**

  Conflict: lectures want students to get marks immediately upon marking. Conversely, the Principal and HoD mandate that no student can see any mark until it has been checked and released by HoD.

  Resolution: HoD policy overrides lecturer preference. The system state machine will lock marks in a “marked” state until the HoD executes a moderation and release workflow, changing the state to “released”.

- **Late work penalties and submission status**

  Conflict: The HoD specifies a hard rule (deducts 10% per day up to 3 days, 0% thereafter), while students argue that late work should not be outright rejected and prefer flexible correction mechanisms.

  Resolution: Implement an automated deduction rule in the system (-10% per calendar day late up to 3 days; submissions on day 4 or later receive 0 marks). The submission is accepted and marked rather than rejected outright, satisfying the students’ desire not to reject late work while enforcing the HoD’s penalty policy.

- **Resubmissions**

  Conflict: Students want to fix mistakes and resubmit before the deadline but the workflow needs clear boundaries so the lecturers are not overwhelmed by infinite resubmissions.

  Resolution: Allow unlimited resubmissions prior to the assignment deadline. Once the deadline passes, regular or late submission rules apply and further resubmission require a lecturer granted extension.

2. Software requirement specification

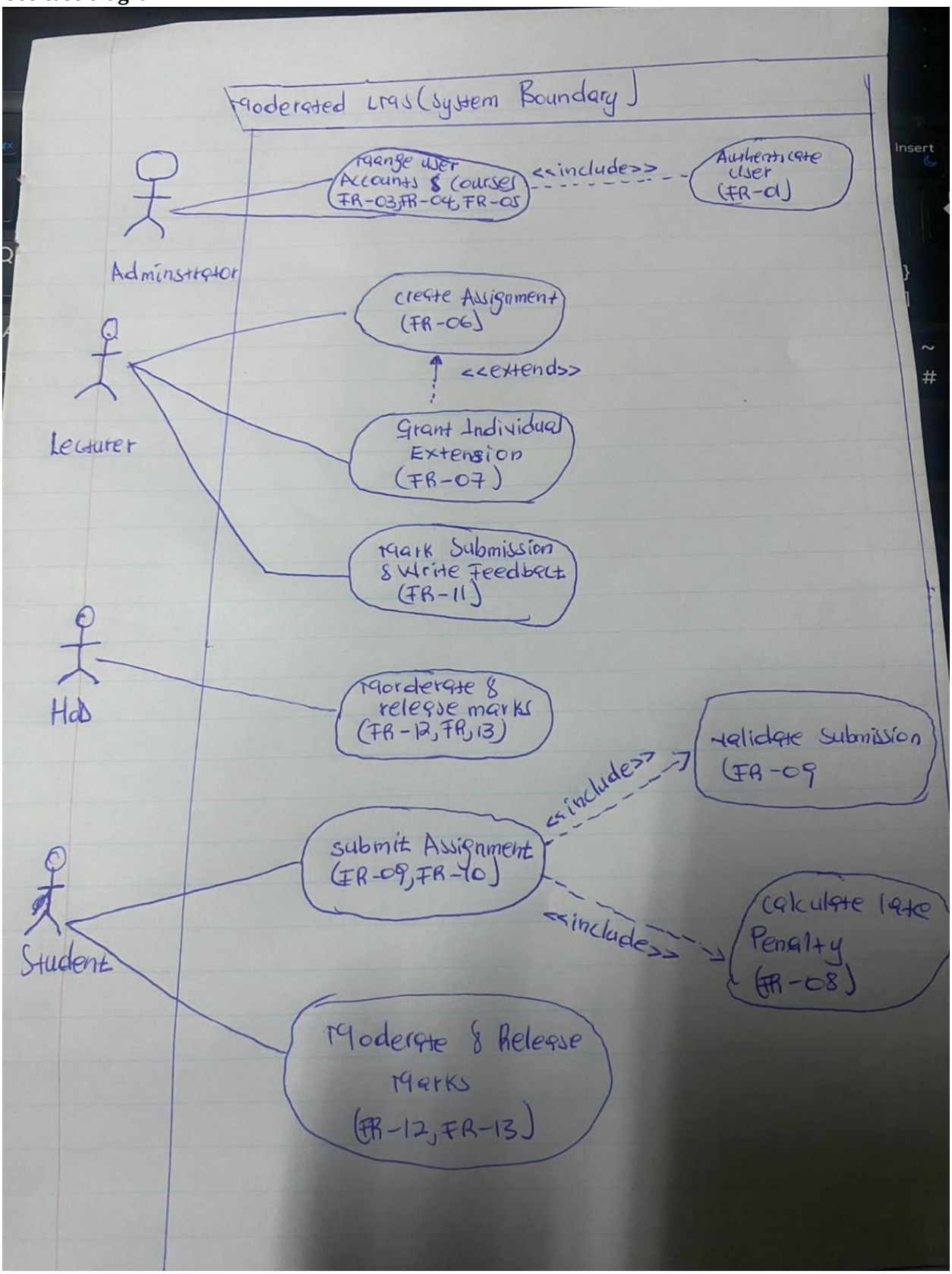
2.1 Functional requirements

- **User authentication:** The system shall authenticate users based on surname and salted password hashes against the SQLite database, restricting access strictly to their assigned role.
- **User and course management:** The Administrator shall be able to create, update and deactivate user accounts, manage department/courses, assign lectures to courses, and enroll students.
- **Assignment submission:** Students shall be able to upload/submit work for their enrolled courses prior to the assignment deadline.
- **Pre-deadline resubmission**: Students shall be able to overwrite or resubmit their assignment files multiple times before the official deadline.
- **Deadline extension:** Lecturers shall be able to grant individual student deadline extensions with a custom new deadline timestamp.
- **Marking and feedback:** Lecturers shall be able to enter numeric marks and type written qualitative feedback for student submissions.
- **Automated late penalty calculation:** The system shall automatically calculate late penalties (deduct 10% per day late up to 3 days, 0 marks afterwards) for submissions received past the effective deadline.
- **HoD review and moderation:** The Head of Department shall be able to view all student submissions, marks and feedback for all courses all within their specific department prior to release.
- **Mark release:** The Head of Department shall be able to approve and bulk release or individually release moderated marks and feedback for all courses within their specific department prior to release**.**
- **Student grade and feedback viewing:** Students shall be able to view their released marks and written feedback for completed assignments.
- **Audit logging:** The system shall record an automated audit log entry for every mark modification, capturing the modifier's ID, timestamp old value, new value and reason.

2.2 Non-functional requirements

- **Performance:** User authentication and dashboard quires should complete within 2 seconds when handling a simulated dataset containing 800 active student records.
- **Security:** Passwords must never be stored in plain text; they should be hashed using a salted cryptographic hashing library.
- **Reliability and office capability:** The application should operate fully offline using a local SQLite database file without requiring an active internet connection.
- **Maintainability:** The C++17 source code should adhere to a documented coding standard and pass static code analysis checks.
- **Testability:** Automated unit tests written using GoogleTest or Catch2 should achieve a minimum of 70% line coverage across the core business logic layer.
- **Portability:** The application should successfully compile via CMake using a standard C++17 compiler on local school desktop environments.

3. **Use case diagram**



- **Use case 1: submit assignment**
  - **Use case ID: UC-C1**
  - **Actor: Student**
  - **Preconditions:**
    - The student is authenticated and logged into the system.
    - The target assignment has been created by a lecturer and belongs to a course the student is enrolled in.
- **Main flow:**
  The student selects an enrolled course and vies the list of active assignments.
  The student selects a specific assignment and uploads their work file/ content.
  The system captures the current system timestamp.
  The system checks the timestamp against the assignment deadline.
  The system stores the file in persistent storage and updates the submission life cycle state to "submitted".
  The system displays a confirmation.
- **Alternative Flows:**
  A1(Resubmission before deadline)
  If the current time is before the deadline, the student uploads a replacement file.
  The system overwrites the previous submission record with the new file and updates the timestamp, maintaining the "submitted" state.

  A2(Late submission)
  If the current time is past the deadline, the system automatically invokes the late penalty calculation rule (-10% per day up to 3 days).
  The system updates the submission life cycle state to "late".
- **Expectation flows:**

  E1(Deadline exceeded past limit): If the submission is received more than three days after the deadline, the system flags the submission as late and assign 0% multiplier according to policy.
  E2(Database failure): If the SQLite database encounters an error during write, the system catches the exception, rolls back the transaction and displays a safe error message without crashing the application.

- **Post conditions:**
  The student's work is securely saved in the database.
  The submission record as many times as they wish before the deadline.

- **Business rule:**
  Students may resubmit work as many times as they wish before the deadline.
  Late work loses 10% of the available mark for each day it is late for up to three days, after that it receives zero.

Use Case 2: Mark Submissions and Provide Feedback

- **Use Case ID**: UC-02
- **Actor**: Lecturer
- **Preconditions**:
    - The lecturer is authenticated and logged into the system.
    - One or more student submissions exist for an assignment within the lecturer's course.
- **Main Flow**:
    - The lecturer accesses their assigned courses and selects a specific assignment to review.
    - The lecturer opens a student's submission file.
    - The lecturer enters a numeric mark (capped at the maximum mark) and writes qualitative feedback text.
    - The lecturer saves the grading record.
    - The system saves the inputs and updates the submission life cycle state to `marked`.
- **Alternative Flows**:
    - **A1 (Granting an individual extension)**:
        1. Before or during grading, the lecturer chooses to grant a student an extension due to extenuating circumstances.
        2. The lecturer inputs a new extended deadline timestamp, which updates the system's evaluation criteria for that student.
- **Exception Flows**:
    - **E1 (Mark exceeds maximum limit)**: If the entered mark is higher than the maximum mark specified for the assignment, the system rejects the input and displays a validation warning.
- **Postconditions**:
    - The mark and written feedback are persistently stored, and the submission moves to the `marked` life cycle state.
- **Business Rules**:
    - Lecturers establish assignment deadlines and maximum marks.
    - Marked submissions remain hidden from students until approved and released by the Head of Department.

Use Case 3: Moderate and Release Marks

- **Use Case ID**: UC-03
- **Actor**: Head of Department (HoD)
- **Preconditions**:
    1. The HoD is authenticated and logged into the system.
    2. Student submissions for courses within the HoD's department have been graded and are in the “marked” state.

- **Main Flow**:

    1. The HoD navigates to the department review dashboard.
    2. The HoD selects a course and reviews student submissions, lecturer marks, and written feedback.
    3. The HoD approves/moderates the marks for the course.
    4. The system updates the submission states from “marked" to “moderated”.
    5. The HoD clicks "Release Approved Marks".
    6. The system updates the submission life cycle state to “released”, making grades visible to students.
- **Alternative Flows**:
    - **A1 (Modifying a mark during moderation)**:
        1. During review, the HoD decides to alter a student's mark.
        2. The HoD inputs the new mark and provides a justification reason.
        3. The system records the adjustment and writes a mandatory audit log entry (capturing who changed the mark, when, old value, new value, and reason).
- **Exception Flows**:
    - **E1 (Attempting to release unmoderated marks)**: If the HoD attempts to release marks while some submissions are still pending or unverified, the system blocks the action and prompts the user to review all records.
- **Post conditions**:
    - All approved course marks transition to the `released` state and can be viewed by students.
    - All alterations are permanently logged in the audit trail.
- **Business Rules**:
    - Students must not see any mark until it has been checked and released by the Head of Department.
    - All changes to marks must be recorded so that accountability is fully maintained.